EX. NO: 2   IMPLEMENTATION OF DOUBLE DATA ENCRYPTION STANDARD (2DES)

AIM

To implement the Double Data Encryption Standard (2DES) encryption and decryption algorithm using Java.

DESCRIPTION

Double Data Encryption Standard (2DES) is an enhancement of the DES algorithm. It applies the DES encryption algorithm twice using two different secret keys. The plaintext is first encrypted using the first key (K1), and the resulting ciphertext is encrypted again using the second key (K2). During decryption, the reverse process is performed using the keys in reverse order. Compared to single DES, 2DES provides improved security against brute-force attacks.

ALGORITHM

STEP-1: Read the plaintext message from the user.

STEP-2: Generate two DES secret keys (K1 and K2).

STEP-3: Encrypt the plaintext using Key K1.

STEP-4: Encrypt the resulting ciphertext using Key K2.

STEP-5: Display the final encrypted message.

STEP-6: Decrypt the ciphertext using Key K2.

STEP-7: Decrypt the resulting data using Key K1.

STEP-8: Display the original plaintext.

CODE

```
import javax.crypto.Cipher;

import javax.crypto.KeyGenerator;

import javax.crypto.SecretKey;

import java.util.Base64;

import java.util.Scanner;


public class DoubleDES {


    public static void main(String[] args) {
```

```
try {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter the message: ");
    String message = sc.nextLine();

    // Generate two DES keys
    KeyGenerator keyGen = KeyGenerator.getInstance("DES");
    SecretKey key1 = keyGen.generateKey();
    SecretKey key2 = keyGen.generateKey();

    // First Encryption
    Cipher cipher1 = Cipher.getInstance("DES");
    cipher1.init(Cipher.ENCRYPT_MODE, key1);
    byte[] firstEncrypt = cipher1.doFinal(message.getBytes());

    // Second Encryption
    Cipher cipher2 = Cipher.getInstance("DES");
    cipher2.init(Cipher.ENCRYPT_MODE, key2);
    byte[] secondEncrypt = cipher2.doFinal(firstEncrypt);

    String encrypted = Base64.getEncoder().encodeToString(secondEncrypt);

    System.out.println("\nEncrypted Message:");
    System.out.println(encrypted);
```

```
            // First Decryption
            cipher2.init(Cipher.DECRYPT_MODE, key2);
            byte[] firstDecrypt = cipher2.doFinal(secondEncrypt);

            // Second Decryption
            cipher1.init(Cipher.DECRYPT_MODE, key1);
            byte[] original = cipher1.doFinal(firstDecrypt);

            System.out.println("\nDecrypted Message:");
            System.out.println(new String(original));

        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Output   Clear

```
Enter the message: running

Encrypted Message:
NoJSH43EkMPPpW1DPsXpUg==

Decrypted Message:
running

=== Code Execution Successful ===
```

RESULT

Thus, the Double Data Encryption Standard (2DES) encryption and decryption algorithm was implemented successfully using Java